Title: STA301 Lecture 01 Comprehensive Study Guide

Subject: Introduction to Statistics (STA301)

Topics:

- Population vs. Sample & Parameter vs. Statistic
- Descriptive vs. Inferential Statistics
- Types of Data & Variables (Discrete vs. Continuous)
- Scales of Measurement (NOIR: Nominal, Ordinal, Interval, Ratio)
- Sources of Data & Statistical Errors (Sampling vs. Non-sampling)

Summary

This study guide consolidates the fundamental principles of introductory statistics covered in STA301 Lecture 01. It covers core statistical definitions, the distinction between working with whole populations versus representative samples, data collection classifications, and measurement scales. Additionally, it highlights key areas identified from practice quizzes—specifically distinguishing parameters from statistics, understanding interval vs. ratio measurement scales, and identifying types of statistical errors.

Key Concepts

1. Population vs. Sample & Parameter vs. Statistic

- **Population:** The complete collection of all elements, individuals, or measurements of interest.
- **Sample:** A smaller, representative subset of a population selected for analysis.
- **Parameter:** A numerical summary measure calculated from an **entire population** (e.g., population mean μ).
- **Statistic:** A numerical summary measure calculated from a **sample** (e.g., sample mean $\bar{x}$).
- *Key Insight:* If data includes **every** individual in a group (e.g., all 15,000 enrolled university students), any calculated average is a **parameter**, not a statistic.

2. Descriptive vs. Inferential Statistics

- **Descriptive Statistics:** Methods concerned with organizing, summarizing, and presenting data using tables, charts, and summary measures (e.g., computing class average GPA).
- **Inferential Statistics:** Methods that use sample results to make inferences, estimates, or conclusions about an entire population (e.g., testing a drug on 500 patients to draw conclusions about all nationwide adults).

3. Variables and Types of Quantitative Data

- **Variable:** A characteristic or attribute that can assume different values across different subjects.
- **Quantitative Discrete Variable:** Numeric data resulting from **counting**; takes distinct, separate values (e.g., number of defective items).
- **Quantitative Continuous Variable:** Numeric data resulting from **measuring**; can take any real value within an interval (e.g., exam completion time, height, weight).

4. Scales of Measurement (NOIR)

- **Nominal:** Data consisting of names, categories, or labels with no natural ranking (e.g., eye color, gender).
- **Ordinal:** Categorical data with a meaningful order/ranking, but differences between values cannot be measured precisely (e.g., satisfaction level: Poor, Fair, Good, Excellent).
- **Interval:** Ordered numerical data with equal intervals between values, but **no absolute (true) zero point** (e.g., Temperature in Celsius or Fahrenheit—where 0°C does not mean complete absence of heat).
- **Ratio:** Ordered numerical data with equal intervals AND a **true absolute zero point** representing the total absence of the attribute (e.g., annual household income, distance, age).

5. Data Sources & Error Types

- **Primary Data:** Original data collected first-hand directly by the researcher for a specific study.
- **Secondary Data:** Pre-existing data compiled or published by another agency prior to the study (e.g., using official government census reports).
- **Sampling Error:** The natural difference between a sample statistic and the true population parameter, occurring solely due to studying a sample rather than the entire population. (A census eliminates sampling error completely).
- **Non-Sampling Error:** Human error or operational flaw during data collection, entry, or processing (e.g., typing an age as 82 instead of 28).

Vocabulary List

- **Statistics:** The branch of science concerned with the collection, organization, presentation, analysis, and interpretation of numerical data.
- **Census:** A complete study or survey covering every single unit in a target population.
- **Parameter:** A numeric metric that describes an entire population.
- **Statistic:** A numeric metric that describes a sample taken from a population.
- **Discrete Data:** Numerical values that are distinct and countable with no intermediate values.
- **Continuous Data:** Numerical values obtained by measurement that can take any value along a continuum.
- **Interval Scale:** Measurement level with ordered, equal increments but an arbitrary zero point.
- **Ratio Scale:** Measurement level featuring an absolute zero point, allowing ratio comparisons (e.g., "double the income").
- **Sampling Error:** Deviations resulting naturally from evaluating a sample instead of the full population.
- **Non-Sampling Error:** Mistakes arising from survey design, faulty entry, non-response, or measurement bias.

Key Questions for Active Recall

1. **Why is the mean GPA of all 15,000 enrolled students at a university considered a parameter rather than a statistic?**
    - *Answer:* Because it is calculated using data from every member of the target population, not just a sample subset.
2. **Why is temperature measured in Celsius classified under the Interval scale instead of the Ratio scale?**

- *Answer:* Because 0°C is an arbitrary zero point (freezing point of water) and does not represent a complete absence of thermal energy.

3. **If a researcher accidentally transcribes a subject's income as $500,000 instead of $50,000, what type of statistical error has occurred?**
   - *Answer:* A non-sampling error (specifically a data entry/processing error).
4. **What is the primary difference between primary data and secondary data?**
   - *Answer:* Primary data is gathered directly by the researcher first-hand for the specific research goal, whereas secondary data was previously gathered and published by someone else.
5. **How does conducting a complete census affect sampling error?**
   - *Answer:* It completely eliminates sampling error because data is collected from every single unit in the population.